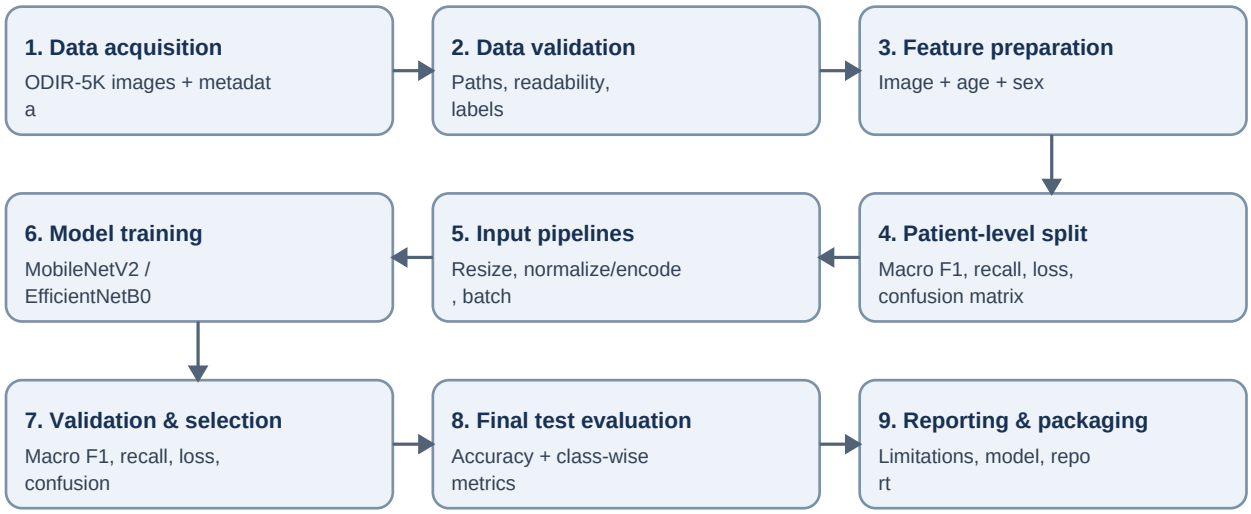
ODIR-5K Project Lifecycle & Pipeline

Multimodal ocular-disease classification using retinal fundus images, age, and sex

1. End-to-end lifecycle overview



Core principle: split by patient before model training. Use the validation set to choose the model; keep the test set untouched until the final evaluation.

2. Detailed stages

Stage 1 — Dataset acquisition

Downloaded the ODIR-5K dataset and located preprocessed fundus images and metadata.

Stage 2 — Validation and inspection

Checked image existence/readability, confirmed image dimensions and RGB format, inspected label distribution and missing values.

Stage 3 — Modeling table

Prepared image-level records with patient ID, filename, image path, label, age, sex, and eye side.

Stage 4 — Leakage-aware splitting

Used patient-grouped train/validation/test splits: 4,478 train images, 948 validation images, and 966 test images. Verified zero patient overlap across splits.

Stage 5 — Clinical preprocessing

Standardized age using training-set statistics only; represented sex with two indicator features. Applied the same transform to validation and test.

Stage 6 — Image preprocessing

MobileNetV2 pipeline: decode JPEG, resize to 224 × 224, apply MobileNetV2 preprocessing. EfficientNetB0 pipeline: decode and resize to 224 × 224, leave pixels in the 0–255 range for the Keras EfficientNet model's internal rescaling.

Stage 7 — Multimodal architecture

CNN image encoder → dropout/dense projection; clinical features → dense/dropout branch; concatenate branches → dense fusion layer → dropout → 8-class softmax.

Stage 8 — Training experiments

Compared full class weights, no class weights, fine-tuning, smoothed class weights, mild augmentation, and EfficientNetB0. Used early stopping and learning-rate reduction callbacks.

Stage 9 — Validation and model selection

Compared validation accuracy, loss, macro F1, per-class recall, and confusion matrices. Selected EfficientNetB0 for final evaluation based on the recorded validation metrics.

Stage 10 — Final test evaluation

Evaluated the selected model once on the held-out test set: 52.48% accuracy, 0.4369 macro F1, 0.507 weighted F1.

Stage 11 — Reporting and handoff

Documented class-wise performance, failure modes, limitations, and the research-only nature of the model.

3. Data split and class imbalance

Split	Images	Unique patients (approx.)	Purpose
Train	4,478	2,350	Fit model weights
Validation	948	504	Tune training and select model
Test	966	504	One-time final evaluation

Class imbalance was addressed experimentally through class weighting and smoothed weights. These methods changed the balance between overall accuracy and minority-class behavior; no single strategy solved all classes.

4. Final model results

Metric	Test result
Accuracy	52.48%
Macro F1	0.4369
Weighted F1	0.507
Strongest recall	Myopia: 94.6%; Cataract: 77.3%
Weakest recall	Hypertension: 4.2%; Other abnormalities: 8.5%

5. Important limitations

This is an educational research prototype, not a clinical tool. The model's low recall for Hypertension and Other abnormalities makes it unsuitable for diagnostic use. Results on ODIR-5K do not establish generalization to other populations, cameras, or clinical settings. Clinical deployment would require independent external validation, clinical oversight, calibration, and safety/regulatory review.

6. Lifecycle completion checklist

✓ Dataset downloaded and validated

✓ Metadata cleaned and target labels prepared

✓ Patient-level split completed and overlap checked

✓ Image and clinical pipelines implemented

✓ Multiple model/training configurations compared

✓ Validation-based model selection completed

✓ Held-out test evaluation completed once

✓ Metrics and limitations documented